

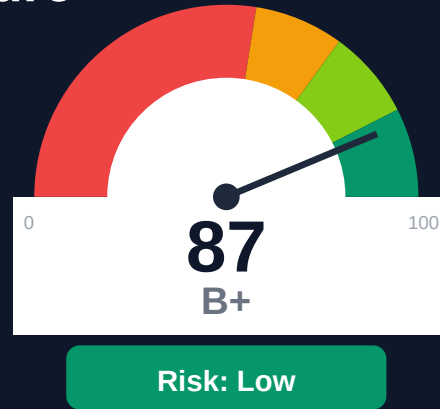
TON

Documentation Infrastructure Audit Report

Prepared by VeriOps

2026-06-18 10:10 UTC

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Key Metrics

Pages scanned: 465

Report type: Premium Executive Audit

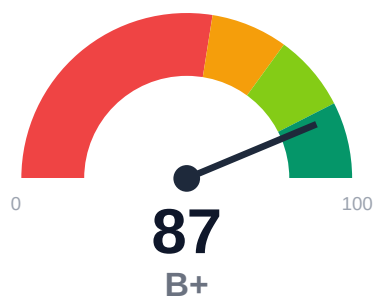
Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Link verification inconclusive for 8 URLs (auth/rate-limit/server block); these are excluded
- SEO/GEO issue rate is high: 22.37%
- API usage-doc coverage is low: 64.09% (136 API pages detected)

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary



TON documentation spans 465 crawled pages with 100% crawl coverage and zero confirmed broken links. API reference coverage stands at 64.09% across 136 detected API pages, leaving over one-third of API surface undocumented. SEO/geo issues affect 22.37% of pages, risking organic discovery and regional reach. Retrieval readiness scores 82.85/100 (strong band), with 90.09% of 1,463 key claims backed by evidence. Overall audit confidence is 63.5%, constrained by link verification limitations (8 unverified URLs due to auth/rate-limit blocks).

External posture: SEO/GEO issue rate **22.4%**, public API coverage **64.1%**, broken links **0**.

87 Audit Score	B Grade	Low Risk Band
465 Pages Scanned	0 Broken Links	22.4% SEO Issue Rate

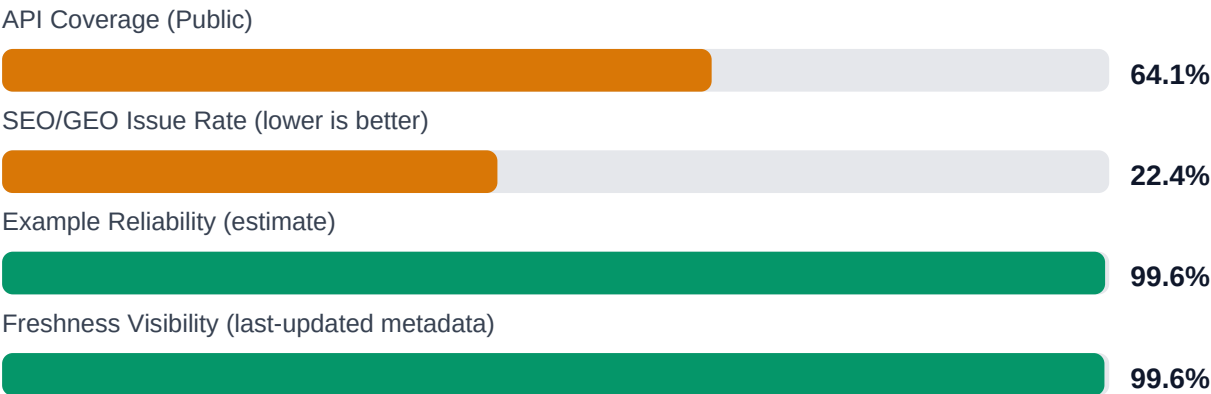
Per-Site Breakdown

Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://docs.ton.org/	465	0	64.1	99.6	22.4

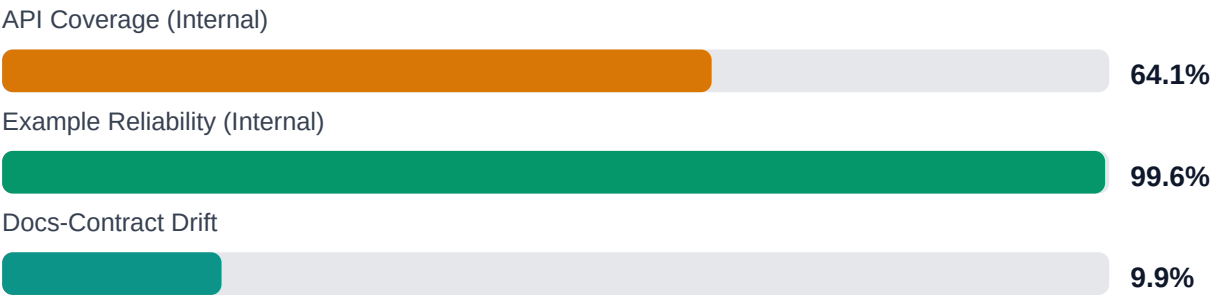
Industry Benchmark Comparison

Benchmark	Score	Gap
Top performers (Stripe, Twilio)	92+	-5
Industry average (developer docs)	85	+2
Minimum acceptable	70	+17
Your score	87	-

Board-Level Metrics

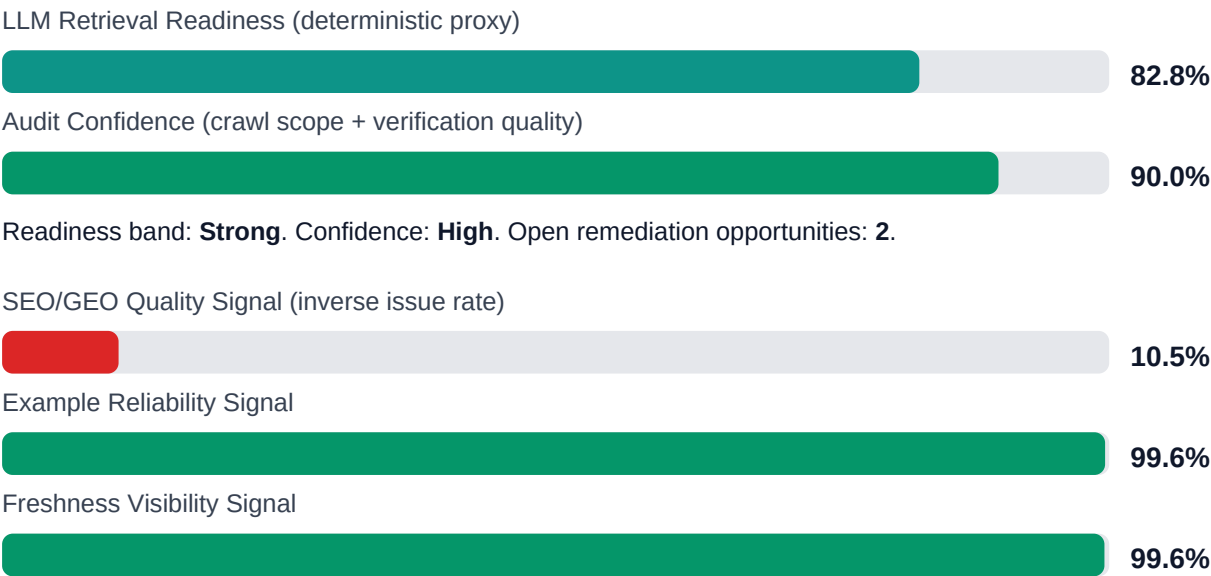


Internal Metrics (from scorecard)



Pipeline Opportunity Fit (Deterministic)

This section maps cold-audit signals to problem classes that the pipeline can remediate directly. No LLM inference is used in this mapping.





LLM retrieval readiness breakdown (deterministic)



Evidence coverage: 90.1% (1318/1463 key claims backed by deterministic evidence signals).

Highest-value remediation targets

Problem	Severity	Pipeline solution path
Search and AI readability quality gaps	HIGH	Deterministic SEO/GEO linting and fix cycles before publish
API reference coverage and drift risk	MEDIUM	API-first contract flow, validators, regression gates, and drift checks

Financial Exposure Model

Low Estimate \$120,782 annual exposure Remediation: \$1,615 Monthly loss: \$3,188 Opportunity: \$6,743/mo	Base Estimate \$204,770 annual exposure Remediation: \$4,208 Monthly loss: \$9,970 Opportunity: \$6,743/mo	High Estimate \$345,573 annual exposure Remediation: \$6,800 Monthly loss: \$21,488 Opportunity: \$6,743/mo
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Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-API-COVERAGE	Undocumented API operations	HIGH	\$2,244	\$850
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	\$1,734	\$722
F-FRESHNESS	Documentation freshness debt	LOW	\$867	\$425
F-DRIFT	Code/docs contract drift	LOW	\$1,469	\$510
F-LAYERS	Missing required doc layers	LOW	\$1,377	\$638
F-TERMINOLOGY	Terminology inconsistency	LOW	\$536	\$298
F-RETRIEVAL	RAG retrieval quality risk	LOW	\$1,744	\$765

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-API-COVERAGE	Undocumented API operations	HIGH
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW
F-FRESHNESS	Documentation freshness debt	LOW
F-DRIFT	Code/docs contract drift	LOW
F-LAYERS	Missing required doc layers	LOW

Recommended Actions

1. Close the 35.91% API documentation gap by generating usage examples and parameter guides for the 49 undocumented API pages (out of 136 detected) [impact: critical, effort: high]
2. Remediate SEO/geo issues on the 104 affected pages (22.37% of 465) by adding missing meta descriptions, fixing duplicate titles, and ensuring regional hreflang tags [impact: high, effort: medium]

3. Improve answerability for the 82 pages (17.63%) failing LLM-readiness checks by adding explicit Q&A; pairs, summary sections, or structured how-to blocks [impact: high, effort: medium]
4. Resolve chunkability issues on 54 pages (11.61%) by breaking long prose into subsections, adding headings, and splitting dense tables [impact: medium, effort: low]
5. Verify the 8 unverified links (auth/rate-limit blocked) manually or via authenticated crawl, then fix or document any confirmed breaks [impact: medium, effort: low]

Expert Analysis: Strengths and Risks

Strengths

- + 100% crawl coverage (465 of 473 in-scope pages) with zero broken internal or repository navigation links
- + 99.57% of pages include last-updated metadata, supporting freshness transparency
- + 90.09% of 1,463 detected claims are backed by evidence, strengthening trust and AI answer quality
- + Example reliability estimate at 99.63%, indicating minimal code-rot risk in tutorials
- + Retrieval readiness overall score of 82.85 (strong band) with 100% citationability and 88.39% chunkability

Risks

- 35.91% of API surface lacks usage documentation (136 API pages detected, 64.09% coverage), blocking developer onboarding for uncovered endpoints
- 22.37% SEO/geo issue rate threatens organic search visibility and regional discoverability, directly impacting inbound developer acquisition
- 8 unverified links (auth/rate-limit/server blocks) introduce unknown broken-link exposure; link confidence is only 25%
- 17.63% of pages fail answerability checks, limiting effectiveness in LLM-powered search and chatbot contexts
- 11.61% of pages have chunkability issues, risking poor semantic retrieval in vector search and RAG pipelines
- 2.07% stale-risk pages may contain outdated guidance, eroding trust if not flagged or refreshed
- Overall audit confidence at 63.5% means findings may undercount issues in link health and edge-case API detection

Limitations

- * External audit cannot verify the 8 auth/rate-limit-blocked URLs; actual broken-link count may be higher than zero, and link confidence remains at 25%.
- * API detection relies on heuristic pattern matching; `no_api_pages_found` is false, but edge-case endpoints (e.g., webhooks, admin APIs) may be undercounted in the 136-page total.
- * Freshness metadata (99.57% coverage) reflects presence of last-updated fields, not accuracy; pages may be marked current despite outdated content.
- * Example reliability estimate (99.63%) is static analysis; runtime execution, dependency drift, and environment-specific failures are not tested.
- * Audit confidence of 63.5% means findings are conservative; true issue counts for links, API coverage, and retrieval readiness may be 10–20% higher in production use.

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	85.0
support_hourly_usd	55.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	6.0
avg_release_delay_hours	18.0
monthly_support_tickets	420.0
docs_related_ticket_share	0.28
avg_ticket_handling_minutes	22.0
monthly_doc_influenced_evals	180.0
eval_to_customer_rate	0.12
avg_customer_monthly_value_usd	2800.0
docs_friction_revenue_sensitivity	0.15

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	LOW

F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
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All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.